MANUFACTURING WORK INSTRUCTION

WC-100 INDUSTRIAL WEAR COMPONENT

FINAL DIMENSIONAL INSPECTION AND DEBURRING

Document Number: WI-WC-100-001
Revision: A
Document Status: Demonstration Document
Document Type: Manufacturing Work Instruction
Product: WC-100 Industrial Wear Component
Process: Final Inspection and Deburring
Prepared By: Technical Documentation
Effective Date: 01 August 2026

DOCUMENT CONTROL

Field	Details
Document Number	WI-WC-100-001
Document Title	WC-100 Final Dimensional Inspection and Deburring
Revision	A
Document Type	Manufacturing Work Instruction
Product	WC-100 Industrial Wear Component
Process	Final Inspection and Deburring
Primary Users	Manufacturing Operators and Quality Inspectors
Status	Demonstration
Effective Date	01 August 2026

---

1. PURPOSE

This work instruction defines the standard method for performing final dimensional inspection and deburring of the WC-100 Industrial Wear Component following machining.

The instruction is intended to ensure that the finished component:

- Meets specified dimensional requirements.
- Is free from unacceptable burrs and sharp edges.
- Has the required product identification.
- Meets the applicable visual quality requirements.
- Is ready for final quality verification and release.

The instruction provides a consistent sequence of activities for manufacturing and inspection personnel.

---

2. SCOPE

This instruction applies to the final inspection and deburring of WC-100 components manufactured according to the applicable product specification and engineering requirements.

The instruction covers:

1. Work area preparation
2. Verification of product identity
3. Review of applicable documentation
4. Visual inspection
5. Dimensional inspection
6. Deburring
7. Final verification
8. Product identification
9. Recording inspection results
10. Handling non-conforming components

This instruction does not replace the approved engineering drawing, product specification, inspection plan or quality procedure.

Where a conflict exists between this work instruction and an approved engineering requirement, the responsible engineering or quality function shall be consulted.

---

3. RESPONSIBILITIES

Manufacturing Operator

The manufacturing operator is responsible for:

- Following the approved work sequence.
- Using the specified tools and equipment.
- Performing required visual checks.
- Completing applicable process records.
- Reporting abnormal conditions or non-conformances.

Quality Inspector

The quality inspector is responsible for:

- Performing or verifying required dimensional inspections.
- Confirming acceptance criteria.
- Reviewing inspection records.
- Supporting disposition of non-conforming products.

Supervisor

The supervisor is responsible for:

- Ensuring personnel are trained or authorized to perform the activity.
- Ensuring required equipment is available.
- Escalating production or quality issues when required.

---

4. SAFETY REQUIREMENTS

Before starting the activity, ensure that the work area is clean, adequately illuminated and free from unnecessary obstructions.

Required personal protective equipment shall be worn in accordance with site safety requirements.

Typical PPE may include:

- Safety glasses
- Protective footwear

- Suitable work gloves when handling components
- Additional PPE required by the site risk assessment

Safety Precautions

1. Handle components using appropriate lifting or handling methods.
2. Do not place hands near sharp or unfinished edges.
3. Use tools only for their intended purpose.
4. Do not use damaged inspection equipment.
5. Keep the work area free from loose metal fragments.
6. Report unsafe conditions immediately.
7. Follow applicable site-specific safety procedures.

---

5. REQUIRED TOOLS AND EQUIPMENT

The following equipment may be required for the activity:

Equipment	Purpose
Vernier Caliper	Overall dimensional measurement
Micrometer	Thickness measurement
Bore Gauge or Suitable Gauge	Hole verification
Steel Rule	General dimensional reference
Deburring Tool	Removal of minor burrs
Abrasive Pad	Edge finishing where approved
Clean Cloth	Cleaning component surfaces
Inspection Record	Recording measurement results

All calibrated measurement equipment shall be within its valid calibration period before use.

---

6. REFERENCE DOCUMENTS

The following controlled documentation should be available before starting the activity:

Document	Reference
Product Specification	SPEC-WC-100
Engineering Drawing	DRG-WC-100
Inspection Plan	IP-WC-100
Quality Procedure	Applicable Site Procedure
Non-Conformance Procedure	Applicable Quality Procedure

The latest approved revision of each applicable document shall be used.

---

7. WORK AREA PREPARATION

Step 1 — Prepare the Work Area

1. Confirm that the inspection area is clean.
2. Ensure adequate lighting is available.
3. Remove unnecessary material from the inspection surface.
4. Confirm that required tools and inspection equipment are available.
5. Verify that measurement equipment is within its calibration period.

Acceptance Check

Proceed only when the work area and required equipment are ready.

---

8. PRODUCT IDENTIFICATION VERIFICATION

Step 2 — Verify Product Identity

Before inspection, confirm the identity of the component.

Verify:

- Product number
- Part number
- Revision
- Batch or heat identification
- Manufacturing identification

Compare the product identification against the applicable controlled documentation.

Acceptance Check

Do not proceed if the product identity or revision cannot be confirmed.

Escalate the issue to the supervisor or quality function.

---

9. VISUAL INSPECTION

Step 3 — Perform Visual Inspection

Inspect the component under adequate lighting.

Check for:

- Cracks
- Significant surface damage
- Excessive burrs
- Sharp projections
- Visible deformation
- Contamination
- Incorrect or missing identification
- Other defects that could affect fit or function

Acceptance Criteria

The component shall be free from defects that could adversely affect:

- Product fit

- Product function
- Product safety
- Product performance
- Subsequent manufacturing operations

Components that do not meet the visual requirements shall be identified and controlled as non-conforming product.

---

10. DIMENSIONAL INSPECTION

Step 4 — Measure Overall Length

Using an appropriate calibrated measuring instrument:

1. Position the component securely.
2. Measure the overall length.
3. Record the measurement.
4. Compare the result with the product specification.

Requirement

Nominal: 250 mm
Tolerance: ±1.0 mm

Acceptance Range

249.0 mm to 251.0 mm

Step 5 — Measure Overall Width

1. Position the component correctly.
2. Measure the overall width.
3. Record the measurement.
4. Compare the result against the specified tolerance.

Requirement

Nominal: 150 mm
Tolerance: ±1.0 mm

Acceptance Range

149.0 mm to 151.0 mm

Step 6 — Measure Component Thickness

Using a suitable calibrated micrometer:

1. Select the required measurement points.
2. Position the measuring instrument correctly.
3. Measure the component thickness.
4. Record the result.

Requirement

Nominal: 25 mm
Tolerance: ±0.5 mm

Acceptance Range

24.5 mm to 25.5 mm

Step 7 — Verify Mounting Hole Diameter

Using an appropriate calibrated measurement method:

1. Verify the mounting hole diameter.
2. Record the measurement.
3. Compare the result against the specified requirement.

Requirement

Nominal: 22 mm
Tolerance: ±0.2 mm

Acceptance Range

21.8 mm to 22.2 mm

Step 8 — Verify Hole Centre Distance

Measure the centre-to-centre distance between the applicable mounting holes.

Requirement

Nominal: 100 mm
Tolerance: ±0.5 mm

Acceptance Range

99.5 mm to 100.5 mm

---

11. DEBURRING

Step 9 — Inspect Edges

Inspect all accessible edges for unacceptable burrs or sharp projections.

Identify areas requiring deburring.

Step 10 — Remove Minor Burrs

Using an approved deburring tool:

1. Secure the component appropriately.
2. Position the tool safely.
3. Remove the burr using controlled movement.
4. Avoid removing excessive material.
5. Inspect the edge after deburring.

Important

Deburring shall not alter critical dimensions or product geometry.

If removal of a defect may affect a specified dimension or functional surface, stop the operation and consult the appropriate quality or engineering function.

---

12. FINAL VISUAL CHECK

Step 11 — Perform Final Visual Verification

After deburring, inspect the component again.

Confirm:

- No unacceptable burrs remain.
- No sharp projections remain.
- No damage has been introduced during deburring.
- Product identification remains legible.
- Component surfaces are reasonably clean.

---

13. FINAL DIMENSIONAL VERIFICATION

Step 12 — Confirm Critical Dimensions

Where required by the inspection plan, repeat critical dimensional measurements after deburring.

Confirm that the component remains within the specified tolerances.

Any measurement outside the acceptance range shall be treated as a non-conformance.

---

14. INSPECTION RECORD

Record inspection results using the applicable inspection record.

Example Inspection Record

Characteristic	Nominal	Tolerance	Actual	Result
Overall Length	250 mm	±1.0 mm	250.2 mm	PASS
Overall Width	150 mm	±1.0 mm	149.8 mm	PASS

Thickness	25 mm	±0.5 mm	25.1 mm	PASS
Hole Diameter	22 mm	±0.2 mm	22.1 mm	PASS
Hole Centre Distance	100 mm	±0.5 mm	100.2 mm	PASS

Actual measurements shown above are demonstration values only.

---

15. PRODUCT ACCEPTANCE

The component may proceed to the next stage when:

1. Required dimensional measurements meet specification.
2. Visual inspection requirements are satisfied.
3. Burrs and sharp projections have been appropriately removed.
4. Product identification is confirmed.
5. Required inspection records are complete.
6. No unresolved non-conformance exists.

---

16. NON-CONFORMING PRODUCT

If any requirement is not met:

1. Stop the affected activity where appropriate.
2. Identify the component as non-conforming.
3. Prevent unintended use or release.
4. Record the non-conformance according to the applicable process.
5. Notify the appropriate supervisor or quality representative.
6. Obtain the required disposition.

Possible dispositions may include:

- Rework
- Repair
- Engineering review
- Use-as-is with appropriate authorization
- Scrap

Personnel shall not independently approve deviations from specified requirements unless authorized by the applicable process.

---

17. TRACEABILITY

Inspection and production records shall maintain traceability to the applicable component or batch where required.

Traceability information may include:

- Product number
- Part number
- Batch number
- Heat number
- Inspection date
- Inspector/operator identification
- Equipment identification
- Inspection result

Records shall be retained according to the applicable quality and document-retention requirements.

---

18. DOCUMENTATION AND RECORD CONTROL

This work instruction is a controlled document.

Users shall verify that they are working from the latest approved revision.

Printed copies may become uncontrolled unless specifically identified as controlled copies.

Changes to this work instruction shall follow the applicable document-review and approval process.

---

19. DOCUMENT REVISION IMPACT

Changes to the product specification or manufacturing process may require this work instruction to be reviewed.

Potential documentation impacts may include:

- Product Specification
- Engineering Drawing
- Inspection Plan
- Manufacturing Work Instruction
- Quality Records
- Training Guide
- Technical Manual
- User Guide

The technical writer or document owner should coordinate with relevant SMEs and process owners to determine documentation impact.

---

20. TRAINING REQUIREMENT

Personnel performing this activity shall receive appropriate training or authorization before independently performing the work.

Training should cover:

- Work instruction requirements
- Applicable product requirements
- Measurement techniques
- Tool usage
- Safety requirements
- Non-conformance handling
- Documentation and record requirements

Training completion shall be recorded according to the applicable training-management process.

---

21. REVISION HISTORY

Revision	Date	Description	Author
A	01-Aug-2026	Initial demonstration release	Technical Documentation

22. PORTFOLIO DEMONSTRATION NOTICE

This work instruction is an independently created technical writing demonstration developed for a professional portfolio.

WC-100 Industrial Wear Component is a fictional product.

All processes, measurements, requirements, inspection values and manufacturing scenarios contained in this document are fictional and have been created solely to demonstrate technical writing and manufacturing documentation practices.

This document does not contain confidential, proprietary or customer information belonging to any employer or third party.

END OF DOCUMENT

Document Number: WI-WC-100-001
Revision: A
Status: Demonstration Document